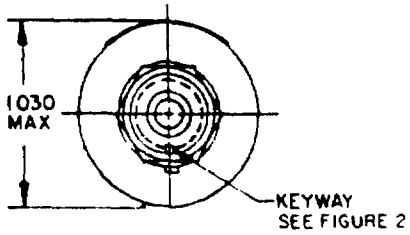
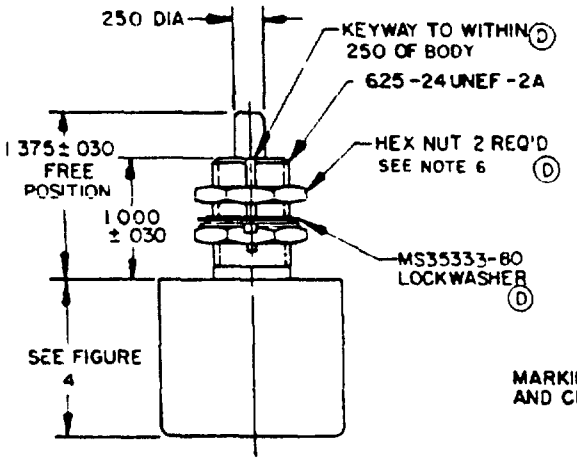
MILITARY SPECIFICATION SHEET

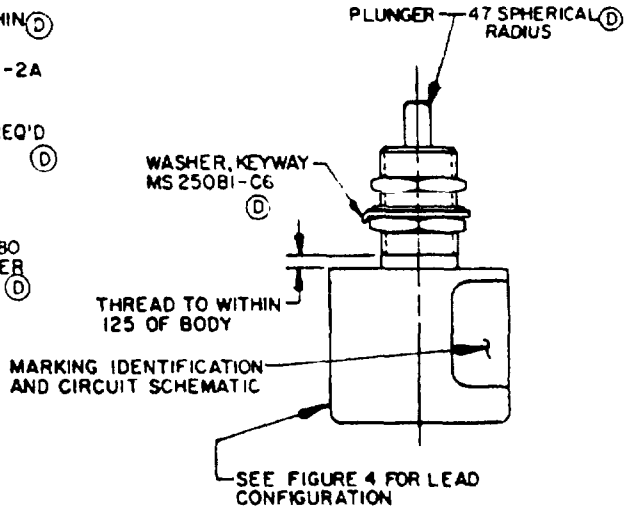
SWITCHES, SENSITIVE, PLUNGER, 10 AMPERES 2PDT AND 7 AMPERES 4PDT, RESILIENT SEAL

The complete requirements for procuring the switches described herein shall consist of this document and the latest issue of Specification MIL-S-8805



INCHES	MM
.030	76
125	3 18
250	6 35
47	11.94
625	15 88
1 000	25 40
1 030	26 16
1 375	34 93





NOTES

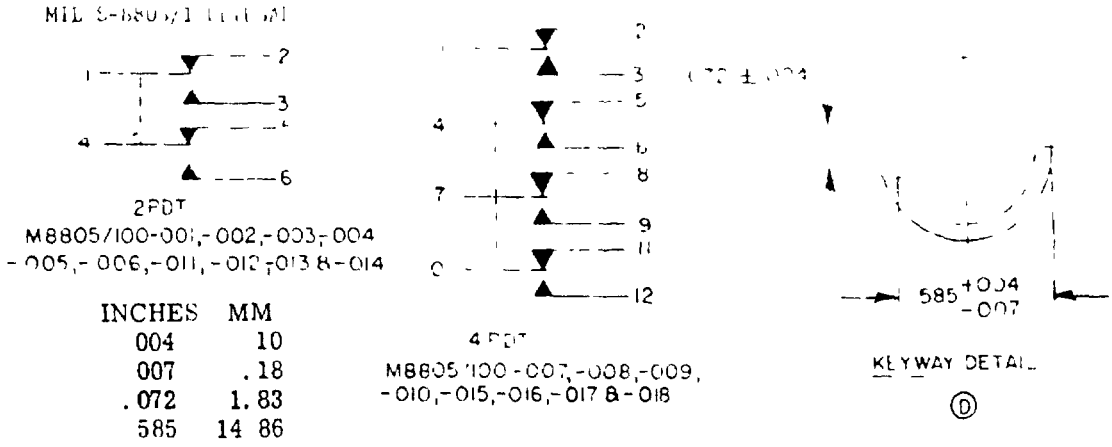
1. Dimensions are in inches
2. Unless otherwise specified, tolerances are ± .010 (25 mm) for two place decimals and ±.005 (13 mm) for three place decimals.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
4. Contour optional provided maximum dimensions specified are not exceeded.
5. The marking (identification and circuit schematic) shall be permanently and legibly marked on the switch case in the location shown (on side of body opposite keyway).

Ⓓ 6. Hex nut shall be MS21340-05 or same except corrosion resisting steel per QQ-S 764-300 series

FIGURE 1. Pin plunger switch

Ⓓ denotes changes

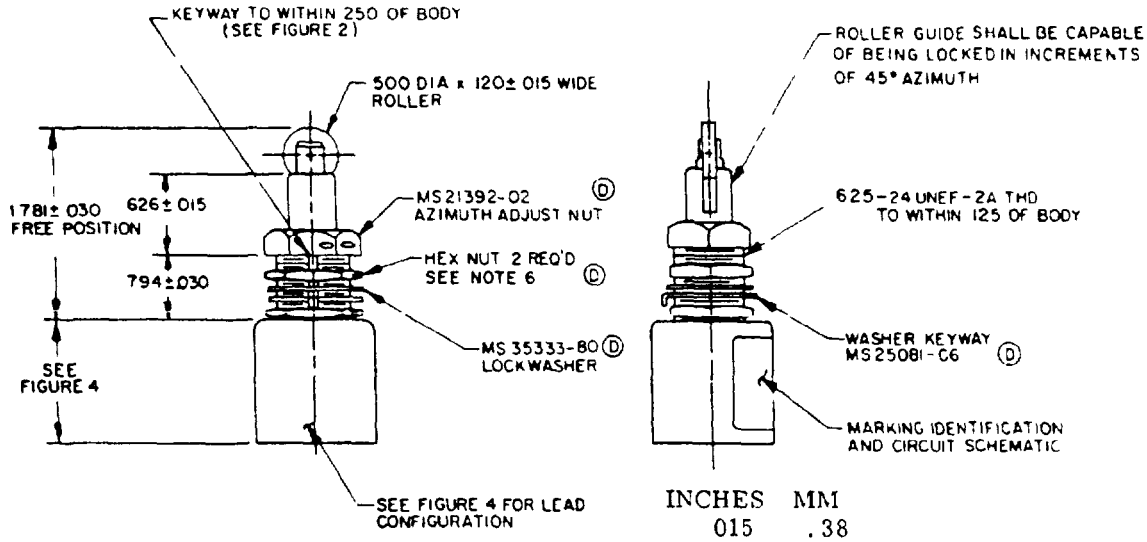
Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.



INCHES	MM
004	10
007	.18
.072	1.83
585	14 86

NOTES
1. Dimensions are in inches
2. Metric equivalents (to the nearest 01 mm) are given for general information only and are based upon 1 inch = 25 4 mm

FIGURE 2 Circuit schematic and keyway details



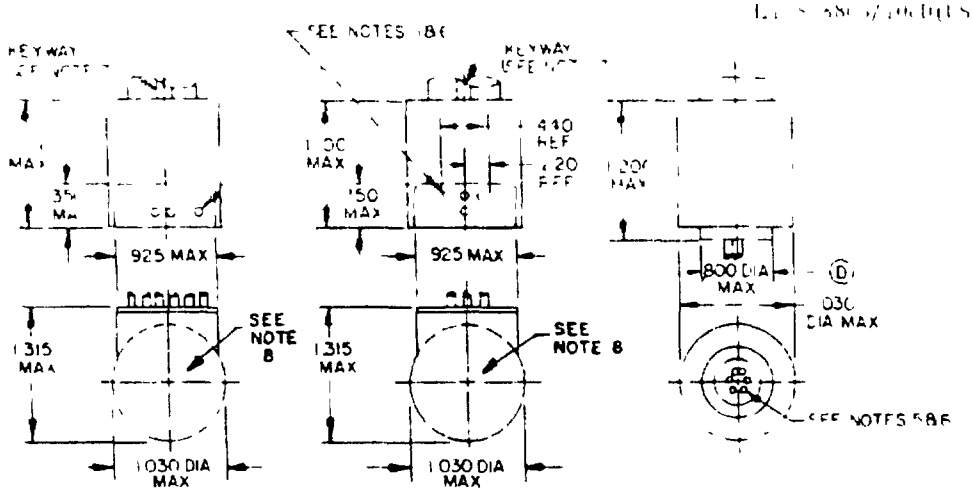
INCHES	MM
015	.38
030	.76
120	3.05
250	6.35
500	12.70
625	15 88
626	15 90
.794	20.17
1.781	45.24

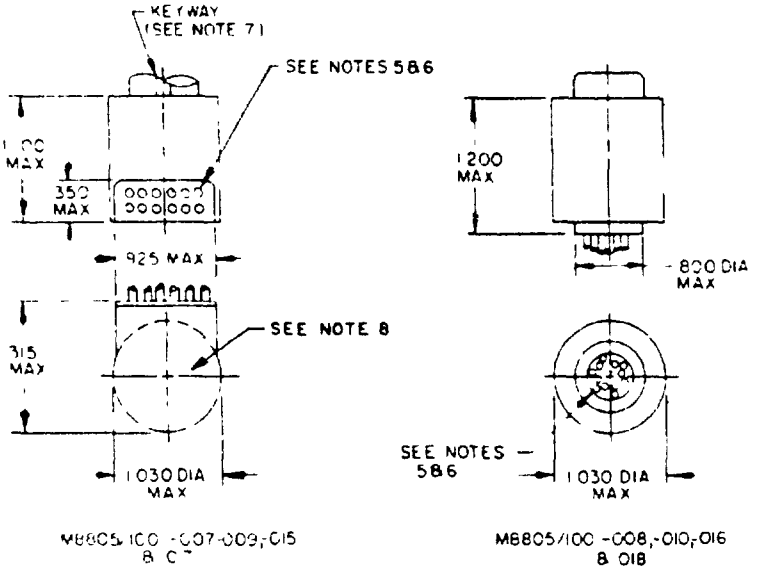
NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 (.25 mm) for two place decimals and ±.005 (13 mm) for three place decimals.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
4. Contour optional provided maximum dimensions specified are not exceeded
5. The marking (identification and circuit schematic) shall be permanently and legibly marked on the switch case in the location shown (on side of body opposite keyway)

Ⓓ 6 Hex nut shall be MS21340-05 or same except corrosion resisting steel per QQ-S-764-300 series

FIGURE 3 Roller plunger switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.





INCHES	MM
.220	5.59
.350	8.89
.440	11.18
.800	20.32
.925	23.50
1.030	26.16
1.100	27.94
1.200	30.48
1.315	33.40

NOTES

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± .010 (.25 mm) for two place decimals and ± .005 (.13 mm) for three place decimals.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Contour optional provided maximum dimensions specified are not exceeded.
5. (D) Wire leads in accordance with M22759/7-20 permanently marked at 3.00 inch (76.20 mm) intervals with switch circuit identification number followed by wire gage number (1-20, 2-20, etc.). Lead length shall be in accordance with table IV.
6. Wire exit shall minimize lead-dress strain transferred to potting.
7. For switches with side exit wires, keyway and wire exit shall be on the same side of switch.
8. (D) Use of mounting holes is optional. If a single hole is used, the largest dimension of this hole shall not exceed .250 inch (6.35 mm). If multiple holes are used, the largest hole dimension shall not exceed .130 inch (3.30 mm).

FIGURE 4. Lead configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.

Number of pins and configuration	See figures 1 and 2
Insulation resistance	1 Resistant) 1/
Temperature characteristics	1 -55° to +85°C continuous duty plus switches shall be subjected to 10 cycles of temperature variations. One cycle is defined as the temperature and times, in order shown, in table I. Immediately upon completion of testing at a temperature requirement, the switches shall be inserted into the subsequent test chamber. At the end of the temperature condition of each cycle, the switches shall make, carry, and break rated resistive current at rated voltage (see 3.6.1) for the number of operations shown in table I. Test shall begin after oven temperature stabilizes. Prior to each make/carry/break operation involving rated resistive current the switch shall carry without making or breaking 10 percent of rated resistive current for 200 milliseconds. Failures, defined as the inability to make or break rated resistive current, shall not occur throughout these tests.

TABLE I. Temperature cycling

Temperature (°C)	Time (minutes)	Number of switch operations
110	30	30
174	10	10
215	1	2

Shock type	Method 213, test condition B (75G), MIL-STD-202
Vibration grade	2 (10 to 2,000 Hz)
Weight	For M8805/100-001 thru -006 .60 pounds maximum For M8805/100-007 thru -010 .80 pounds maximum For M8805/100-011 thru -014 .74 pound maximum For M8805/100-015 thru -018 1.09 pounds maximum
Operating characteristics	
Actuating force	9 ±3 pounds
Full overtravel force	30 pounds maximum
Releasing force	5 pounds minimum
Pretravel	.070 inch maximum
Movement differential	.035 inch maximum
Overtravel	.250 inch minimum.
Coincidence of operating and releasing points	All circuits shall transfer within .010 inch of plunger travel after first circuit transfers.
Strength of actuating means	100 pounds minimum
Azimuth adjust nut	No damage to switch or azimuth adjust nut shall result from torquing this nut to 15 pound-inches minimum
Material	
Roller	Corrosion resistant steel QQ-S-764, type 416, condition B, 38-43 Rc. Finish to be bright, scale free.
Potting material	MIL-S-23586, type 1, class 2, grade B.

1/ All entrances to the switch cavity, except through the actuator bushing, shall be sealed by fusion of glass to metal or ceramic to metal, and the lead wires shall be potted to provide stress relief.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.

(D) Finish:	All external surfaces of switch housing, including threaded bushing and bracket, shall be both corrosion resistant and electrically conductive. Cadmium plating shall not be used.
(D) Contact resistance	Low level circuit requirements apply.
Dielectric withstanding voltage	
(D) Sea level:	1,250 Vrms.
Altitude	600 Vrms at 80,000 feet. After electrical endurance, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable.
Mechanical endurance·	25,000 cycles.
Electrical endurance.	25,000 cycles.
(D) Low level circuit.	1,000 cycles at 85°C. (Applies provided switch has not made nor broken more than 250 milliamperes at 6 volts dc or peak ac).
Electrical ratings·	See table II.
(D) Moisture resistance.	When switches are tested wet, the insulation resistance shall not be less than 100 megohms.
(D) Icing.	Applicable.
(D) Case and plunger grounding for EMI·	All case parts shall be bonded together (such as by staking, soldering, welding, or brazing) either along the entire abutting surfaces or at a minimum of four approximately equally spaced areas along each abutting surface. The electrical resistance between the plunger and the wire exit portion of the switch housing bracket most remote from the threaded bushing shall not exceed 100 milliohms under all the following conditions (a) Measurements made in accordance with MIL-STD-202, method 311 (b) Without polishing or cleaning the areas on the switch to which the test leads are attached. (c) With the plunger both fully extended and fully depressed. (d) Throughout the total life of the switch
(D) Marking	Marking shall remain legible after the following test· (a) Soak for a minimum of 72 hours at room temperature in each of the fluids specified below. (b) After each soak period the specimen shall be wiped dry and then rubbed with a dry thumb a minimum of 20 times applying moderate pressure. NOTE. Each test specimen shall be subjected to soak periods in all the fluids. (c) If metal foil identification plates are used, they shall comply with MIL-P-19834 In addition to the legibility requirement defined herein, the identification plate shall not become loosened or partially or completely detached from the switch housing during or after the soak tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.

(d) Lubricating oil, test fluid: Referee
MIL-H-5606, Hydraulic Fluid, Petroleum Base; Aircraft, Missile and Ordnance
MIL-L-7808, Lubricating Oil, Aircraft Turbine Engine, Synthetic Base

Part numbers — M8805/100 - (dash number from table IV)

TABLE II. Electrical rating

Type	Sea level 28 Vdc		80,000 ft 28 Vdc	
	-001 thru -006 and -011 thru -014	-007 thru -010 and -015 thru -018	-001 thru -006 and -011 thru -014	-007 thru -010 and -015 thru -018
	Amperes	Amperes	Amperes	Amperes
Resistive	10	7	10	7
Inductive	4	2	4	2
Motor	6	4	6	4
Lamp	3	2	2	1

Qualification and group C retention of qualification
Group submission — See table III
Group A inspection
Seal test — Only watertight test shall be performed in group A.

Ⓓ TABLE III Qualification and group C tests (group submission)

Examination or test	Samples	Extent of approval
Qualification inspection table of MIL-S-8805 1/ 4/	M8805/100-017 (32 units)	All
Visual and mechanical examination Dielectric withstanding voltage Operating characteristics	M8805/100-016 (2 units)	
Visual and mechanical examination Temperature cycling 2/ (in accordance with table I)	M8805/100-012 (2 units)	
Marking test 3/	M8805/100-017 (2 units)	

1/ Case and plunger grounding test to be accomplished before and after mechanical endurance
2/ Applicable in qualification testing only.
3/ Following immersion, the switch shall also be subjected to terminal strength, insulation resistance, dielectric withstanding voltage, operating characteristics, and seal tests
4/ Group C Explosion test not applicable

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.

Ⓓ TABLE IV Dash numbers and configuration

Dash numbers		Actuator	Lead wires
36 $^{+2}_{-0}$ inch 1/ lead wires	72 $^{+2}_{-0}$ inch lead wires		
-001	---	Pin plunger	Six wire leads brought out in single row.
-002	-011	Pin plunger	Six wire leads brought out in two rows of three leads each.
-003	-012	Pin plunger	Six wire leads brought out from the bottom of the switch.
-004	---	Roller plunger	Six wire leads brought out in single row.
-005	-013	Roller plunger	Six wire leads brought out in two rows of three leads each.
-006	-014	Roller plunger	Six wire leads brought out from the bottom of the switch.
-007	-015	Pin plunger	Twelve wire leads brought out in two rows of six each
-008	-016	Pin plunger	Twelve wire leads brought out from bottom of switch
-009	-017	Roller plunger	Twelve wire leads brought out in two rows of six each.
-010	-018	Roller plunger	Twelve wire leads brought out from bottom of switch.

1/ Inactive for new design. Use dash numbers -011 thru -018.

Custodian
Air Force - 11

Preparing activity:
Air Force - 11

Agent.
DSA - ES

(Project 5930-F436)

U.S. GOVERNMENT PRINTING OFFICE 1976— 703-020 798

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.

FOLD

ASD/ENESS
Wright-Patterson AFB, O 45433

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S. MAIL

ASD/ENESS
Wright-Patterson AFB, Ohio 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes" in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED. 1 OF 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T11:21Z
Check the source to verify that this is the current version before use.